

Task one

BCE

$$\sigma = -\frac{1}{m} \sum_{i=1}^m \log h(x_i)$$

$$(1 - \sigma) \log(1 - h(x))$$

$$\frac{\partial \sigma}{\partial \theta} = -\frac{1}{m} \sum_{i=1}^m \frac{\partial \log h(x_i)}{\partial \theta}$$

$$\frac{\partial \sigma}{\partial \theta} = -\frac{1}{m} \sum_{i=1}^m \left(\frac{1}{h(x_i)} \cdot \frac{\partial h(x_i)}{\partial \theta} \right)$$

$$= -\frac{1}{m} \sum_{i=1}^m \frac{h(x_i)}{h(x_i)(1 - h(x_i))}$$

$$h = \frac{1}{1+e^{-2z}} = (1+e^{-2z})^{-1}$$

$$\frac{\partial h}{\partial z} = 1(1+e^{-2z})^{-2} \cdot e^{-2z} (+1)$$

$$= (1+e^{-2z})^{-2} \cdot e^{-2z}$$

$$= \frac{e^{-2z}}{(1+e^{-2z})^2}$$

$$1-h = 1 - \frac{1}{1+e^{-2z}} = \frac{1+e^{-2z} - 1}{1+e^{-2z}}$$

$$(1-h) = \frac{e^{-2z}}{(1+e^{-2z})^2}$$

(2)

$$(a) \quad \frac{\partial \theta_i}{\partial \omega} = \theta_i - \omega \quad \sum_{i=0}^{\infty} \theta_i x_i + \dots$$

$$\frac{\partial \theta_i}{\partial \omega} = \left(\frac{\partial_i}{\partial_i + \frac{1-i}{\partial_i}} \right) \cdot \frac{1}{\partial_i} \cdot x_i$$

$$\left(\frac{\partial_i}{\partial_i + (1-i)h} \right) x_i$$

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